

6. INTRODUCTION TO THYRISTORS, TRIACS, DIACS

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There are several thyristors displayed on 6.1. Triacs look the same, while diacs look like small power rectifying diodes. Their symbols, and pin-out is found in figure 6.2.

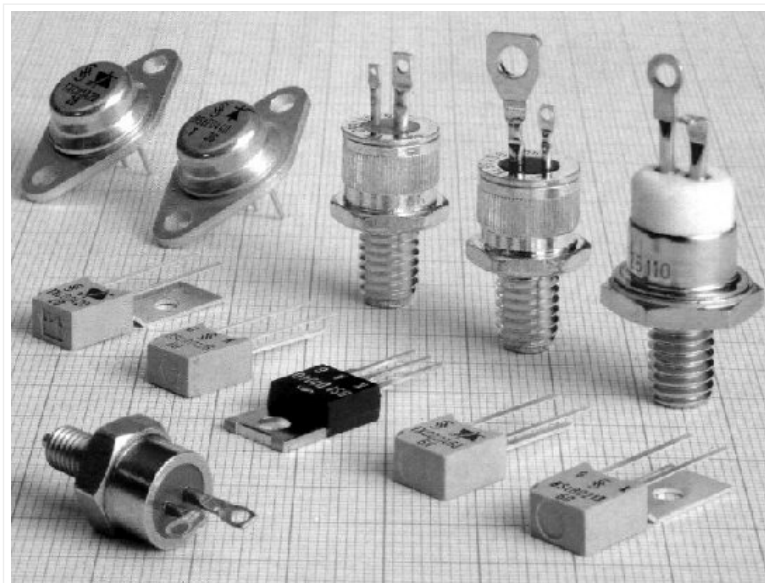


Fig. 6.1: Several thyristors and triacs

A thyristor is an improved diode. Besides anode (A) and cathode (K) it has another lead which is commonly described as a gate (G), as found on picture 6.2a. The same way a diode does, a thyristor conducts current when the anode is positive compared to the cathode, but only if the voltage on the gate is positive and sufficient current is flowing into the gate to turn on the device. When a thyristor starts conducting current into the gate is of no importance and thyristor can only be switched off by removing the current between anode and cathode. For example, see figure 6.3. If S1 is closed, the thyristor will not conduct, and the globe will not light. If S2 is closed for a very short time, the globe will illuminate. To turn off the globe, S1 must be opened. Thyristors are marked in some circuits as SCR, which is an acronym for Silicon Controlled Rectifier.

A triac is very similar to a thyristor, with the difference that it can conduct in both directions. It has three electrodes, called anode 1 (A1), anode 2 (A2), and gate (G). It is used for regulation of alternating current circuits. Devices such as hand drills or globes can be controlled with a triac.

Thyristors and triacs are marked alphanumerically, KT430, for example.

Low power thyristors and triacs are packed in same housings as transistors, but high power devices have a completely different housing. These are shown in figure 6.1. Pin-outs of some common thyristors and triacs are shown in 6.2 a and b.

Diacs (6.2c), or two-way diodes as they are often referred to, are used together with thyristors and triacs. Their main property is that their resistance is very large until voltage on their ends exceeds some predefined value. When the voltage is under this value, a diac responds as a large value resistor, and when voltage rises it acts as a low value resistor.

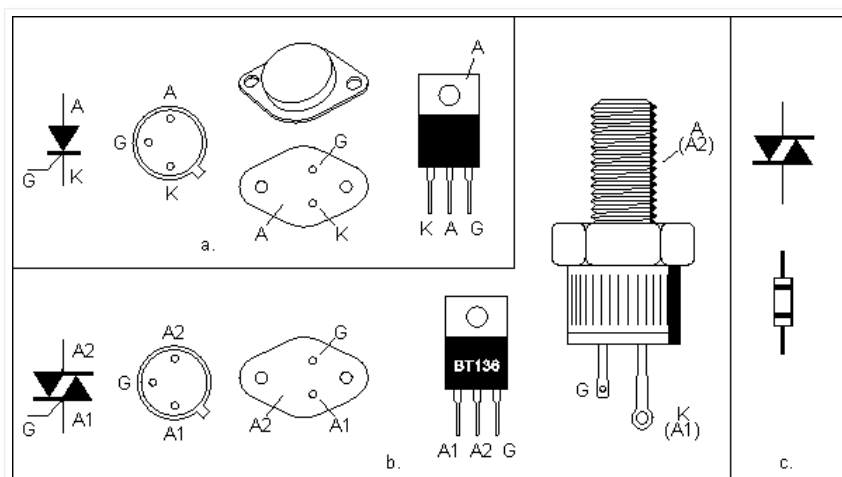


Fig. 6.2: Symbols and pin placements for: a – thyristor, b – triac, c – diac

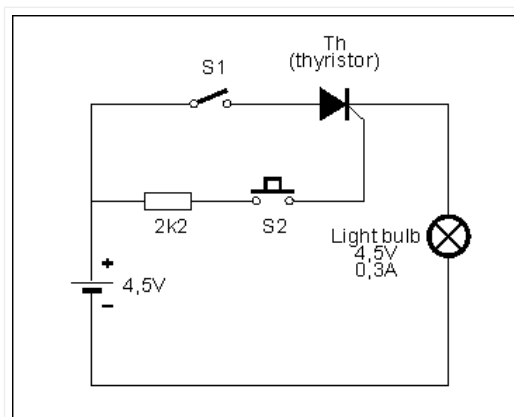


Fig. 6.3: Thyristor principle of work



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